

YenCare

“Healthcare access, wherever you are.” · Team: Product 4

The problem

Many patients in Ghana have limited access to smartphones, mobile data, or reliable internet. Even patients who can book appointments online still lose hours of their day sitting in a waiting room with no idea how long it will take. Clinic staff, meanwhile, manage bookings, walk-ins, and queues largely by hand.

The pitch

YenCare is a clinic appointment and queue system built around one idea: a patient shouldn't need a smartphone to get healthcare access. Everyone books, gets confirmed, and checks their place in line through the channel they already have — a web app for smartphone users, SMS for everyone else, with USSD as a stretch goal once the core is solid. Staff get one dashboard to manage bookings, check-ins, and the live queue.

The differentiator we're pitching on: low-connectivity access combined with a real virtual queue, not just a booking form.

MVP scope for this programme

The original concept (full AWS serverless architecture, Terraform, automatic rescheduling engine, offline sync, full analytics) is a multi-month production build. Scoped honestly against 5 weeks, a Week 3 feature freeze, and a team where three of five members are working from starter templates, here is what we're committing to versus what we're naming as future vision:

MVP — build this	Stretch / future vision — do not commit to gates
Web app: registration, booking, lookup, cancel, reschedule	USSD interface (pitch as vision; only attempt after Gate 3 if time allows)
SMS: booking confirmation + one reminder	Full notification retry/delivery-tracking system
Staff dashboard: check-in, queue status, appointment management	Offline-first sync for the dashboard
Simple virtual queue: Booked → Checked In → Waiting → Called → Completed	Automatic rescheduling engine for doctor unavailability
Queue position + rough wait estimate (simple average, not historical modelling)	Full analytics suite (busiest hours, trend charts)
Roles: Admin, Doctor, Receptionist — basic access control	Full audit log of every staff action
One SMS provider (sandbox), one database, one backend service	AWS Lambda / DynamoDB / EventBridge / SQS / Terraform
Booking clash prevention (DB-level constraint)	Full concurrency test suite — keep to a targeted test, not exhaustive coverage

Tech stack (kept deliberately simple)

- Backend: Node.js / Express REST API
- Database: PostgreSQL (relational fits appointments, patients, and queue entries naturally)
- Frontend: React web app

- SMS: one provider’s sandbox/test tier (e.g. Africa’s Talking or Hubtel) — verified working in Week 1 before we commit to it
- Hosting: a single simple deployment (e.g. Render/Railway or a basic AWS EC2 + S3/CloudFront setup) — no multi-service AWS architecture
- CI: GitHub Actions running tests on push — no Terraform, no multi-environment infrastructure

Milestones mapped to programme gates

Gate	Programme milestone	YenCare deliverable
1 · Fri 4 Sep	Plan approved, scaffold in place	Repo + scaffold live, DB schema drafted, SMS sandbox account working, staff login skeleton
2 · Fri 11 Sep	Core feature working end-to-end	A patient can book an appointment on the web app and receive an SMS confirmation, end to end
3 · Fri 18 Sep	Feature-complete — then freeze	All MVP features working: booking, cancel/reschedule, staff dashboard, check-in, queue position, reminders
4 · Fri 25 Sep	Reviewed, fixed, documented	Bug fixes, race-condition test on booking, README + setup docs, no new features
5 · Fri 2 Oct	Demo delivered, insights shared	Live demo: web booking + SMS flow + staff dashboard + queue check

Team & ownership

Name	Role	Owns
Blessing Edmund Kwame Dogbe	Team Lead · DevOps	SMS/USSD provider integration, hosting & deployment, environment config, weekly Part A submission
Able Kafu Azanda	Builder · Backend	Appointment & queue API, booking concurrency handling, database schema
Emmanuella Lodonu	Foundations · Backend	Patient registration & lookup endpoint, notification-trigger logic
Harry “Ephraim” Nartey	Foundations · Fullstack	Staff dashboard: check-in, queue status view, appointment status changes
Raymond B. Afrani	Foundations · Frontend	Patient-facing web app: booking form, appointment lookup, queue-position display

Why this is worth greenlighting

- It solves a real, specific problem — not a generic CRUD app — for a partner clinic in Ghana.
- The MVP is genuinely achievable by Gate 3, with clear cut lines so feature freeze doesn’t catch us short.
- It plays to the team’s actual strengths: three backend/fullstack members and a DevOps lead, with a realistic, contained frontend scope for our one dedicated frontend Foundations member.
- The vision scales — USSD, offline sync, and analytics are natural next phases if the clinic partner wants to continue after handover.